

JEE MAIN REVISION MASTER: HYDROCARBONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 166 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Hydrocarbons — Applied Cases, Reactions & Traps	114 PYQs	Applied Topics
Very High	Hydrocarbons — Core Principles & Governing Laws	51 PYQs	Core Principle
Standard	Hydrocarbons — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Hydrocarbons — Applied Cases, Reactions & Traps	$(1^{\circ}), \quad \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 \xrightarrow{\text{Cl}_2/\text{H}_2\text{O}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 \xrightarrow{\text{AgNO}_3} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Ag}$
Hydrocarbons — Core Principles & Governing Laws	O_3 , $\text{C}_{10}\text{H}_{14}$, $\text{CH}_3\text{COCl}/\text{AlCl}_3$ will give
Hydrocarbons — Standard Calculations & Problem Archetypes	____, g mol^{-1} H:1, C:12, O:16, Br:80

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hydrocarbons — Applied Cases, Reactions & Traps (Very High Yield — 114 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Hydrocarbons.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The major product of which of the following reaction is not obtained by rearrangement reaction?..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The dibromo compound [P] (molecular formula: $C_{(9)}H_{(10)}Br_{(2)}$) when heated with excess sodamide followed by treatment with dilute HCl ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "Consider the following reaction of benzene. In compound (Q), the percentage of oxygen is ____ %. (Nearest integer)..." [Ans: 10]

3.2 Hydrocarbons — Core Principles & Governing Laws (Very High Yield — 51 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Hydrocarbons.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Hydrocarbons.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Hydrocarbons PYQs.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Identify the correct IUPAC name of hydrocarbon (x) containing three primary carbon atoms and with molar mass 72g mol^{-1} " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Arrange the following alkenes in decreasing order of stability. Choose the correct answer from the options given below:..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "Given below are two statements : Statement I : Benzene is nitrated to give nitrobenzene, which on further treatment $\text{with } CH_3COCl/A...$ " [Ans: N/A]

3.3 Hydrocarbons — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Hydrocarbons.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (Numerical): "The cycloalkene (X) on bromination consumes one mole of bromine per mole of (X) and gives the product (Y) in which C:Br ratio is 3:1..." [Ans: 66]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 23rd January Evening Shift (Q#1)

Consider the following reaction of benzene. In compound (Q), the percentage of oxygen is ____ %. (Nearest integer)

Correct Answer: 10

Solution Analysis: Molecular mass of compound Q is 157. In Q, oxygen contributes mass 16 (for one oxygen atom). So, percentage of oxygen in Q is % of oxygen in product 'Q' is $= (16 / 157) \times 100 = 10.19\%$ Nearest integer = 10 %.

Benchmark Problem 2: JEE Main 2026 (Online) 22nd January Morning Shift (Q#2)

The cycloalkene (X) on bromination consumes one mole of bromine per mole of (X) and gives the product (Y) in which C:Br ratio is 3:1. The percentage of bromine in the product (Y) is ____ %. (Nearest integer) (Given : molar mass in gmol^{-1} : H:1, C:12, O:16, Br:80)

Correct Answer: 66

Solution Analysis: Bromination of a cycloalkene (one double bond) consumes 1 mole of Br_2 per mole of alkene, which means 2 bromine atoms add across the double bond. So, product (Y) is a dibromo compound. Given in (Y), the ratio C:Br=3:1. Let number of carbon atoms = n. Number of bromine atoms = 2. So, $(n / 2) = (3 / 1) \Rightarrow n = 6$ Hence (Y) has formula: $\text{C}_6\text{H}_{10}\text{Br}_2$ (since cyclohexene is C_6H_{10} and it adds Br_2) Now molar mass of (Y) : Carbon: $6 \times 12 = 72$ Hydrogen: $10 \times 1 = 10$ Bromine: $2 \times 80 = 160$ Total molar mass: $72 + 10 + 160 = 242 \text{ gmol}^{-1}$ Perc

Benchmark Problem 3: JEE Main 2026 (Online) 6th April Morning Shift (Q#3)

Given below are two statements : Statement I : Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of CH_3MgBr with water. Statement II : Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction. In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true **[Correct]**

Correct Answer: D

Solution Analysis: Analysis of Statement I Statement I says: "Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of CH_3MgBr with water." Decarboxylation of sodium ethanoate: This reaction involves heating a sodium salt of a carboxylic acid with soda-lime (a mixture of NaOH and CaO). The general reaction is the removal of a carboxyl group. For sodium ethanoate (CH_3COONa), the reaction is: $\text{CH}_3\text{COONa} + \text{NaOH} \rightarrow \text{CH}_4 + \text{Na}_2\text{CO}_3$ As you can see, methane (CH_4) is indeed the product. So, this